

ESLSCA BUSINESS SCHOOL PARIS

MBA - Trading & Financial Markets

**To what extent does Reinforcement Learning
reduce market impact costs (Implementation
Shortfall) on fragmented European equity
markets?**

Research Dissertation

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All empirical results in this dissertation are simulated Implementation Shortfall under a transparent, literature-calibrated market model. No proprietary or real tick data is used; only signs, rankings and comparative statics are interpreted. The full pipeline is reproducible from the accompanying code repository.

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Abstract

This dissertation investigates the extent to which Reinforcement Learning (RL) can reduce market-impact costs, measured by the Implementation Shortfall (IS), in the fragmented post-MiFID II European equity market. It frames multi-venue optimal execution as a Markov Decision Process and compares two RL agents - a value-based, discrete-action agent (DQN family) and a policy-gradient, continuous-action agent (PPO family) - against the standard TWAP, VWAP and Almgren-Chriss benchmarks. Because synchronised multi-venue Level-2 data for CAC 40 equities is available only through paid institutional feeds, no proprietary or real tick data is used: the agents are trained and evaluated inside a stochastic market simulator calibrated to publicly documented CAC 40 characteristics, with market-impact coefficients from the published literature (Almgren et al., 2005). This calibrated-simulation approach is standard in the RL-execution literature (Cartea, Donnelly & Jaimungal, 2018; Hendricks & Wilcox, 2014).

On a held-out set of 12,000 out-of-sample sessions, the PPO-family agent reduces mean simulated IS by 33.8% relative to TWAP, 31.7% relative to VWAP and 34.9% relative to an optimally-calibrated Almgren-Chriss (all paired, Newey-West $p < 0.001$). The DQN-family agent achieves 29.0% / 26.8% / 30.2% and is significantly outperformed by PPO (6.7%, $p < 0.001$), consistent with H1a. Multi-venue routing generates a fragmentation alpha that rises with cross-venue price dispersion and is exactly zero when dispersion vanishes - a falsifiable confirmation of H2 - reaching approximately 35% IS reduction at a moderate 8 bp dispersion. The RL advantage is strongly regime-conditional (H3): the PPO-versus-Almgren-Chriss gain rises monotonically from 12.6% in calm markets ($\sigma = 15\%$) to 67.6% in stressed markets ($\sigma = 35\%$); contrary to the original sub-hypothesis H3b, a small but statistically significant edge persists even in calm regimes. The ranking is sign-stable under square-root impact, Student-t shocks, higher impact, and an out-of-distribution parameter shift.

All figures are simulated IS under stated assumptions; only signs, rankings and comparative statics are interpreted - they are not a backtest on, or a claim of deployable edge in, the live market. The full pipeline is openly reproducible from the accompanying code repository.

***Keywords:** Reinforcement Learning, Implementation Shortfall, MiFID II, optimal execution, Markov Decision Process, Smart Order Routing, DQN, PPO, CAC 40, market simulation.

General Introduction

Practical and Sectoral Context

The efficient execution of large institutional orders constitutes one of the most critical operational challenges for participants in financial markets. When a portfolio manager decides to buy or sell a significant position - whether for portfolio rebalancing, directional trading, index replication, or constrained divestment - the very act of executing that order on the market moves prices against the manager's interest. This phenomenon, known as market impact, represents a substantial implicit cost that, combined with explicit brokerage commissions and transaction taxes, forms the total cost of transaction.

The measurement and minimisation of this execution cost have become major strategic imperatives for asset managers, pension funds, insurance companies, and hedge funds operating in European capital markets. The economic stakes are substantial. The Banque de France (2022) estimated that market impact costs represent on average 15 to 40 basis points depending on order size and stock liquidity for institutional orders on European markets. To illustrate the financial significance concretely: a fund managing €10 billion with 50% annual portfolio turnover and a mean IS of 20 bp bears an annual execution cost approaching €10 million - a figure that could be reduced by several million euros through the adoption of optimised execution strategies. For large pension funds managing hundreds of billions in assets, the aggregate annual savings from superior execution can reach tens of millions of euros, directly translating into improved net returns for beneficiaries.

Beyond this direct financial impact, execution quality has gained visibility in the wake of regulatory developments under MiFID II. The obligation for investment firms to demonstrate best execution to their clients - monitoring, documenting, and regularly reviewing execution quality across all execution venues - has transformed what was once an operational back-office concern into a strategic front-office priority. Asset managers now face direct accountability to institutional clients for the execution quality of their orders, making the optimisation of execution algorithms a competitive differentiator in the European asset management landscape.

The context in which these execution decisions must be made has been profoundly transformed over the past two decades. MiFID I, effective November 2007, abolished the national concentration principle and unleashed pan-European inter-venue competition. MiFID II, effective January 2018, further deepened this fragmentation. Today, according to ESMA data (2023), liquidity in a CAC 40 stock is distributed in real time across eight to twelve simultaneous active venues, with market shares that evolve dynamically in response to macroeconomic announcements, the strategies of electronic

market makers, and regulatory constraints such as the Double Volume Cap on dark pool trading. This fragmentation, while theoretically improving competition for small retail orders, considerably complicates execution decisions for large institutional orders that cannot be filled at a single venue without significant adverse price impact.

The Limits of Traditional Approaches

Traditional execution algorithms - Time-Weighted Average Price (TWAP) and Volume-Weighted Average Price (VWAP) - were designed for single-market environments and operate on fixed, pre-programmed schedules that are blind to real-time market conditions. TWAP distributes orders uniformly over time, making no attempt to exploit variations in intraday liquidity. VWAP improves on this by following historical volume patterns, but remains vulnerable to days with atypical volume profiles - precisely the high-stakes trading days with major corporate events or macroeconomic releases. Neither algorithm is designed to exploit the cross-venue liquidity opportunities created by MiFID II fragmentation.

More sophisticated analytical models, notably Almgren and Chriss (2001), provide an optimal trajectory that balances market impact against timing risk, but do so within a single-venue, stationary-parameters framework. When deployed on real fragmented European markets, these models must be recalibrated periodically and cannot adapt in real time to regime changes, liquidity crises, or sudden shifts in cross-venue liquidity distribution. The fundamental limitation of all these approaches is their inability to learn from market data: they apply fixed rules, however well-calibrated, to a market environment that is inherently non-stationary and multi-dimensional.

This structural gap between the capabilities of existing tools and the demands of the contemporary European trading environment defines the research opportunity that this dissertation addresses. Reinforcement Learning, which learns adaptive decision policies directly from data through repeated interaction with an environment, offers a conceptually compelling alternative to fixed-rule execution algorithms - provided it can be made to work reliably in the noisy, non-stationary, and high-stakes context of institutional equity execution.

Scientific Context and Research Gap

From an academic perspective, the application of Reinforcement Learning to optimal execution has attracted growing research attention since Nevmyvaka, Feng & Kearns (2006) demonstrated that a tabular Q-learning agent could outperform TWAP on NYSE data by 12-18%. More recent contributions have applied Deep RL - notably DQN (Mnih et al., 2015) and PPO (Schulman et al., 2017) - to increasingly sophisticated execution settings, incorporating

dynamic order book signals (Cartea et al., 2018), multi-asset correlation (Ning et al., 2021), and adversarial market maker dynamics (Spooner et al., 2018).

However, a systematic review of this literature reveals a consistent and significant gap: virtually all published empirical studies on RL-based execution focus on US markets (NYSE, NASDAQ) or stylised simulated environments. The specific context of fragmented European equity markets post-MiFID II - with its distributed liquidity across ten or more heterogeneous venues, its regulatory best execution constraints, its specific adverse selection dynamics, and its distinct intraday liquidity seasonality patterns - has not been systematically addressed. This dissertation fills this gap by conducting a rigorous empirical study using a calibrated stochastic market simulator with CAC 40-like parameters, explicitly incorporating the multi-venue allocation dimension as a first-class component of the RL agent's decision space.

Personal Motivation and Professional Context

The choice of this research question is grounded in professional practice as much as academic interest. Working in a quantitative trading role at Tikehau Capital, a European alternative asset manager, the author is directly confronted with the operational challenges of institutional execution on fragmented European equity markets. The development of backtesting infrastructure and systematic data pipelines for algorithmic strategies provides direct exposure to the data quality, model calibration, and performance attribution challenges that this dissertation addresses theoretically and empirically. This practitioner perspective motivates both the practical orientation of the research question and the managerial focus of the recommendations developed in Chapter 4.

The academic context of the ESLSCA MBA programme, combining financial engineering fundamentals with strategic management frameworks, provides the appropriate setting for research that bridges the quantitative and managerial dimensions of the execution optimisation problem. The dissertation supervisor, Abdelbaric EL ALAOUI, brings the academic rigour and methodological guidance necessary to translate this practitioner motivation into a theoretically grounded and empirically rigorous research contribution.

Research Question

To what extent does Reinforcement Learning reduce market impact costs (Implementation Shortfall) on fragmented European equity markets?

This central question is operationalised through three subsidiary research questions structuring our empirical approach:

- **RQ1.** To what extent does an RL agent (DQN or PPO) trained on a calibrated market simulator outperform traditional execution strategies (TWAP, VWAP, Almgren-Chriss) in terms of out-of-sample Implementation

Shortfall?

- ***RQ2:*** To what extent does multi-venue fragmentation of European markets constitute an exploitable source of execution alpha for an RL agent with dynamic Smart Order Routing capability, beyond simple temporal optimisation?
- ***RQ3:*** Are the IS gains achieved by RL agents conditional on market regime
 - specifically more pronounced during periods of high realised volatility or low visible liquidity?

Scope and Delimitations

This research focuses on intraday equity execution in CAC 40 constituents on European markets, over horizons of 15 to 60 minutes, within the post-MiFID II regulatory framework. We restrict the analysis to buy-side equity orders; results are theoretically transposable to the sell side with minor adjustments. The multi-asset execution problem, simultaneous execution of correlated positions, and derivatives markets are acknowledged as important extensions but excluded from the current scope for tractability. The simulation environment is calibrated to reproduce the statistical properties observed in CAC 40 markets over 2022-2024 - a period that encompasses both relatively calm conditions (2023) and episodes of elevated volatility (2022 rate cycle, 2024 election-driven volatility) - providing a sufficiently diverse market environment for robust generalisation.

Several important caveats bound the interpretation of our results. Our backtesting framework, however carefully constructed, cannot perfectly replicate the feedback dynamics of live trading, particularly the second-order effects of the agent's own impact on other market participants' behaviour. Parameter uncertainty in the market simulator introduces a degree of model risk that is irreducible with backtesting alone. We acknowledge these limitations explicitly and incorporate them into our recommendations for real-world deployment.

Dissertation Structure

This dissertation is structured in two parts. Part 1 establishes the theoretical and conceptual foundations. Chapter 1 surveys the academic literature on Implementation Shortfall, optimal execution models, European market fragmentation, and Reinforcement Learning in finance, identifying the gap that motivates the research. Chapter 2 develops the original conceptual framework, formalises the execution problem as an MDP, and derives testable research hypotheses. Part 2 is devoted to empirical verification. Chapter 3 details the epistemological positioning, simulator design and calibration, and evaluation protocol. Chapter 4 presents and analyses the empirical results, compares them with the literature, and formulates actionable managerial

recommendations. A general conclusion synthesises contributions, limitations, and research perspectives.

Part 1 - Literature Review and Conceptual Framework

The first part of this dissertation grounds our research in the existing academic field and constructs the conceptual framework guiding our empirical investigation. Chapter 1 provides a critical survey of the three pillars of our research: Implementation Shortfall theory, optimal execution models and their limitations, fragmented European market microstructure, and Reinforcement Learning in finance. Chapter 2 translates these theoretical insights into an original conceptual framework, formally models the multi-venue execution problem as a Markov Decision Process, and derives the three testable research hypotheses structuring Part 2.

Chapter 1: Literature Review

Chapter Introduction

This chapter conducts a structured critical review of the academic literature across four axes. Section 1.1 traces the theoretical development of Implementation Shortfall, from Perold's (1988) foundational paper to the multi-venue decompositions of Lehalle & Laruelle (2013), examining the conceptual framework and empirical evidence on execution cost decomposition. Section 1.2 analyses classical optimal execution models, Almgren-Chriss (2001), TWAP, VWAP and their stochastic extensions, their assumptions, optimality conditions, and well-documented limitations. Section 1.3 examines the specific microstructure of fragmented European equity markets post-MiFID II, its genesis, its quantitative characteristics, and its implications for institutional execution. Section 1.4 reviews the theoretical foundations of Reinforcement Learning and its applications to optimal execution and related financial decision problems. Section 1.5 synthesises these four axes to identify the precise research gap that motivates this dissertation.

1.1 Implementation Shortfall: Theoretical Foundations

1.1.1 Perold's Foundational Contribution (1988)

The formalisation of Implementation Shortfall as a measure of execution performance owes its origins to André Perold's 1988 article 'The Implementation Shortfall: Paper Versus Reality', published in the *Journal of Portfolio Management*. Writing in the context of the late 1980s, when institutional equity trading was dominated by block trades on national exchanges and transaction cost measurement was rudimentary, Perold identified a systematic and largely unmeasured divergence between the theoretical performance of portfolio managers and their realised performance. He attributed this divergence to what he termed the implementation shortfall, the gap between the performance of a 'paper portfolio' valued at decision prices and the performance of the 'real portfolio' valued at actual execution prices.

Perold's insight was both empirical and conceptual. Empirically, he documented that this gap was economically significant and systematic, not merely a random noise around the decision price, but a persistent drag on performance attributable to the mechanics of order execution. Conceptually, he argued that this gap represented a first-order performance attributor, comparable in importance to stock selection alpha, that had been largely ignored in the portfolio management literature because it was difficult to measure and attributed to the 'operations' side of the business rather than the

'investment' side. This conceptual reframing had profound implications for the industry, ultimately spawning the Transaction Cost Analysis (TCA) industry and the algorithmic execution business.

Formally, Implementation Shortfall for a buy transaction of target size N shares is defined as:

where n denotes the quantity executed at price S at time t , S_0 is the decision price (mid-market price at the time the investment decision is made), S_{end} is the closing mid-market price of the execution window, and commissions represent all explicit transaction costs. The three terms capture distinct cost components: the realised price cost (difference between execution prices and decision price for filled quantities), the opportunity cost (foregone gain for unfilled quantities, penalising incomplete execution at adverse end prices), and explicit transaction costs. This decomposition reveals the fundamental trade-off of optimal execution: executing slowly to minimize direct price impact increases the opportunity cost and timing risk; executing quickly reduces opportunity cost but increases direct impact.

Figure 1: Implementation Shortfall decomposition following Perold (1988) and Kissell & Malamut (2006)

The practical significance of the IS framework stems from its comprehensive treatment of execution cost. Unlike simpler measures such as the bid-ask spread, which captures only the cost of immediate small-size execution, IS captures the full cost profile of a large order executed progressively over time, including both the direct market impact of the execution process itself and the timing risk borne by the manager during the execution period. This makes IS the measure of choice for institutional investors evaluating execution quality on large orders, and the natural objective function for execution optimisation algorithms.

1.1.2 Microstructural Decomposition of Market Impact

The decomposition of market impact into temporary and permanent components, essential for both modelling and attribution purposes, has been extensively studied in the market microstructure literature. Hasbrouck (1991), building on the information-theoretic framework of Glosten & Milgrom (1985) and Kyle (1985), provided the foundational econometric decomposition distinguishing the two components through their persistence properties in transaction price series.

Temporary market impact, also called price impact or price pressure, corresponds to the price displacement induced by a large order that is subsequently reversed as the order book replenishes. It reflects the mechanical imbalance of supply and demand created by the institutional order's one-sided pressure on the order book: market makers facing a

sustained buy flow adjust their quotes upward to manage inventory risk, but subsequently revert prices toward equilibrium as new natural sellers arrive. Empirically, temporary impact is typically modelled as a function of the instantaneous execution rate v normalised by the average daily volume V , with a power-law relationship:

where η is the temporary impact coefficient and $\delta \approx 0.5-0.6$ is the power-law exponent, calibrated empirically (Almgren et al., 2005). The square-root relationship ($\delta \approx 0.5$), observed by Bouchaud et al. (2010) across a wide range of asset classes and market conditions, is one of the most robust empirical regularities in market microstructure.

Permanent impact, by contrast, reflects the information content of the order. Glosten & Milgrom (1985) demonstrated that in a market populated by both informed and uninformed traders, a rational market maker must set prices to break even in expectation against the possibility that the counterparty is informed. Any buy order, even if it originates from an uninformed liquidity trader, carries some probability of being from an informed trader who has superior information about the stock's fundamental value. This adverse selection risk leads to a permanent price adjustment following large orders, which does not revert because it reflects a genuine update of the market's estimate of fundamental value.

Kyle (1985) formalised this mechanism in a continuous equilibrium model, showing that the price impact of a large order, measured by Kyle's lambda λ , is the market's rational response to updating beliefs about fundamental value conditional on observed order flow. The key insight is that lambda is larger in illiquid markets (where order flow is more likely to be informed) and smaller in liquid markets (where uninformed trading dominates). In the multi-venue context, venue-specific lambdas reflect venue-specific adverse selection environments, dark pools, for instance, tend to attract less informed flow and thus exhibit lower permanent impact per unit of volume.

1.1.3 Contemporary Refinements and Multi-Venue Extensions

Several important refinements to IS decomposition have been developed to address the increasingly complex execution environment. Kissell & Malamut (2006) proposed a three-component decomposition adding timing cost, the IS component attributable to directional price movement during the execution period unrelated to the order's own impact, to market impact and spread cost. This decomposition enables more granular performance attribution: timing cost reflects the market's overall directional movement during execution (alpha or anti-alpha of execution timing), while market impact reflects the order's own price pressure. Understanding the relative magnitudes of these components guides algorithm design: when timing cost dominates (e.g., in

trending markets with strong momentum), faster execution that sacrifices some market impact efficiency may be preferable.

Almgren et al. (2005) conducted a large-scale empirical study estimating IS components across thousands of institutional equity transactions in US markets, finding that temporary impact represents approximately 60% of total impact cost on average, with significant cross-sectional variation driven by stock liquidity and order size. Their calibration methodology, regressing execution cost components on order characteristics and market conditions, has become the industry standard for IS decomposition and provides the empirical foundation for our impact parameter calibration (Table 2).

Lehalle & Laruelle (2013) extended the IS framework to the multi-venue context, proposing a decomposition that isolates the contribution of cross-venue allocation decisions from overall execution performance. Their framework distinguishes a consolidated IS (measured against the consolidated best bid-offer across all venues) from venue-level IIS components reflecting the quality of routing decisions. This multi-venue decomposition is directly relevant to our research: it provides the methodological basis for testing H2 (the fragmentation alpha hypothesis) by allowing us to attribute a fraction of total IS reduction specifically to cross-venue allocation optimization.

1.2 Classical Optimal Execution Models

1.2.1 The Almgren-Chriss Model (2001)

The Almgren and Chriss (2001) model represents the culmination of a decade of research on optimal execution initiated by Bertsimas & Lo (1998) and constitutes the dominant analytical framework for institutional execution algorithm design. Published in the Journal of Risk and rapidly adopted by execution desks at major investment banks, the model derives a closed-form optimal execution trajectory for a risk-averse agent liquidating a large position over a fixed horizon under linear market impact assumptions.

The model setup is as follows. An agent holds an initial position X_0 to liquidate over N discrete time steps of duration $\tau = T/N$. At each step k , the agent chooses n_k shares to liquidate. The mid-price evolves as a random walk: $S_k = S_{k-1} + \sigma\sqrt{\tau} \xi_k$ where σ is annualised volatility and ξ_k is a standard normal shock. The execution price for a liquidation of n_k shares at step k is:

where $\eta \times (n_k/\tau)$ is the temporary impact (proportional to instantaneous execution rate n_k/τ) and $\kappa \times (n_k/\tau) \times \tau = \kappa \times n_k$ is the permanent impact, linearly adjusting the mid-price for all subsequent periods. The agent minimises a mean-variance objective over the Implementation Shortfall: $J(\pi) = E[IS] + \lambda \times \text{Var}[IS]$, where λ is the risk aversion parameter. Almgren and Chriss (2001) derive a closed-form solution showing the optimal number of shares remaining at time t_k follows:

Figure 2: Optimal execution trajectories under Almgren-Chriss (2001) for different values of risk aversion λ ; larger λ leads to faster front-loading

This solution reveals three important qualitative properties. First, the optimal strategy always maintains a non-negative inventory (no short-selling during liquidation). Second, the strategy is front-loaded when risk aversion is high (the agent prefers price certainty over impact minimisation) and uniform (TWAP) when risk aversion approaches zero. Third, the optimal strategy depends on a single composite parameter $\tilde{\kappa}$ that combines risk aversion, volatility, and impact parameters, suggesting that the full parameter space collapses to a one-dimensional family of strategies. This parsimony has made the Almgren-Chriss model extremely practical for implementation and calibration.

1.2.2 Limitations of the Almgren-Chriss Framework

Despite its elegance and widespread industrial adoption, the Almgren-Chriss model rests on several simplifying assumptions that are increasingly inadequate for modern European market execution. Forsyth, Kennedy, Tse & Ware (2012) provided a rigorous analysis of the consequences of non-linear impact, demonstrating that the linear temporary impact assumption leads to suboptimal trajectories when actual impact follows a square-root law. Specifically, they show that under square-root impact, the optimal strategy is more front-loaded than Almgren-Chriss predicts, the agent should execute faster early to take advantage of lower proportional impact at smaller execution rates. The magnitude of this mis-specification bias grows with order size, with Forsyth et al. (2012) estimating a 5-15% suboptimality for orders exceeding 1% of daily volume.

Cartea & Jaimungal (2015) demonstrated that the fixed-horizon assumption, while computationally convenient, is inconsistent with the observed urgency dynamics of institutional execution. In practice, managers often face soft deadlines where early completion is rewarded but late completion is penalised, creating a time-varying risk profile that the static Almgren-Chriss framework cannot capture. Their extension incorporating stochastic urgency significantly alters the optimal trajectory for orders with uncertain completion horizons.

The single-market assumption is perhaps the most problematic for our research context. By design, the Almgren-Chriss model considers a single order book with a single impact function. In the fragmented European market environment, the optimal execution must simultaneously consider eight to twelve venue-specific impact functions, cross-venue correlation structures, and dynamic liquidity distributions, a dimensionality that far exceeds the scope of analytical closed-form solutions and motivates the learning-based approach of this dissertation.

1.2.3 TWAP, VWAP and Their Limitations

The Time-Weighted Average Price (TWAP) algorithm, despite its apparent simplicity, embodies an important insight: by spreading execution uniformly over time, it minimises sensitivity to any specific execution time and achieves a price close to the time-average of market prices over the execution window. For a risk-neutral investor who believes the market is a martingale, TWAP is asymptotically optimal among fixed-schedule algorithms. Its MiFID II auditability, any execution reviewer can verify that trades were approximately uniformly distributed over the scheduled window, makes it the compliance default for many regulated asset managers.

However, TWAP's uniform distribution is also its primary limitation. By ignoring the well-documented U-shaped intraday liquidity pattern, where volumes are highest at the open and close and lowest at mid-session, TWAP systematically executes a disproportionate share of its volume during low-liquidity periods, incurring higher per-unit impact than necessary. Nevmyvaka et al. (2006) demonstrated that even a simple rule-based algorithm that avoids the low-liquidity mid-session hours achieves IS reductions of 5-8% relative to TWAP on NYSE data.

VWAP addresses this by conditioning the execution schedule on the estimated historical intraday volume profile, concentrating execution during high-volume periods where per-unit impact is lowest. Berkowitz, Logue & Noser (1988) formalised this approach and showed empirically that VWAP achieves lower IS than TWAP in normal market conditions by 3-6 bp on average for large-cap US stocks. However, Bialkowski, Darolles & Le Fol (2008) documented a critical failure mode: on days with atypical volume profiles, earnings announcements, central bank meetings, index rebalancing days, the historical volume pattern is a poor predictor of the actual intraday distribution, and the VWAP algorithm's rigid adherence to the historical profile leads to execution concentrated at suboptimal times. These are precisely the high-impact days where execution quality matters most.

1.3 Fragmentation of European Equity Markets

1.3.1 The MiFID Regulatory Framework and Market Architecture

The transformation of European equity market structure from nationally concentrated to pan-European fragmented is one of the most consequential regulatory developments in the history of financial markets. To appreciate its full significance, it is necessary to understand the institutional architecture that MiFID replaced. Before November 2007, equity markets in continental Europe were organised around national 'home' exchanges operating under the concentration rule, a regulatory principle requiring or strongly incentivising the routing of domestic equity orders to the national regulated market. The

Paris Bourse (subsequently Euronext Paris), Frankfurt's Xetra (Deutsche Börse), and the London Stock Exchange enjoyed quasi-monopoly positions in their domestic markets, with minimal competition and correspondingly high transaction costs that were essentially borne by institutional investors and their clients.

MiFID I (Directive 2004/39/EC, effective 1 November 2007) abolished the concentration rule and replaced it with the Best Execution principle: investment firms were required to obtain the best possible result for their clients on an ongoing basis, considering price, costs, speed, likelihood of execution, and size. This single regulatory change created the conditions for pan-European inter-venue competition, and new entrants moved rapidly to exploit the opportunity. Chi-X Europe, backed by a consortium of major broker-dealers, launched in 2007 offering fees up to 10 times lower than Euronext and a matching engine with sub-millisecond latency. Within two years, Chi-X had captured 15-25% of volumes in major European equities, permanently ending the national exchange monopolies. BATS Europe and Turquoise followed, further atomising liquidity.

MiFID II (Directive 2014/65/EU, effective 3 January 2018) addressed the unintended consequences of the original fragmentation while deepening it in other dimensions. The Double Volume Cap (DVC) mechanism limited dark pool trading to 4% per venue and 8% aggregate of total volumes in any individual stock over any rolling 12-month period, diverting previously dark volumes toward lit venues. The creation of the Organised Trading Facility (OTF) category institutionalised new forms of multilateral trading for non-equity instruments. The enhanced transparency requirements, particularly the post-trade reporting obligations under RTS 10 and the best execution reporting under RTS 27/28, created a data infrastructure enabling systematic cross-venue performance comparison. These changes reconfigured the competitive dynamics among venues, shifting flows between lit and dark, between regulated markets and MTFs.

Figure 3: Liquidity distribution by venue for a representative CAC 40 stock (TotalEnergies, Q2 2024), Source: ESMA (2023)

1.3.2 Quantitative Characteristics of European Fragmentation

ESMA's Annual Statistical Report on EU Securities Markets (2023) provides the most comprehensive and authoritative data on the current state of European equity market fragmentation. For stocks in the major European indices (CAC 40, DAX 40, FTSE 100, AEX), the report documents a consistent pattern: Euronext Paris or the relevant national exchange retains 30-40% of visible lit volume as the primary venue of price discovery, while 40-50% of volumes are distributed across three to four major MTFs (CBOE Europe,

Turquoise, Aquis Exchange, and Euronext's pan-European trading services), with the remaining 10-20% traded in various dark formats (broker crossing systems, systematic internalisers, dark MTFs under DVC limits).

This aggregate picture, while informative, understates the dynamic complexity that execution algorithms must navigate. The cross-venue distribution of liquidity is highly time-varying at multiple frequencies. On an intraday basis, the opening and closing auctions on regulated markets concentrate 20-30% of daily volumes in brief periods where MTFs are inactive or operate in distinct modes, creating sharp transitions in the optimal routing strategy between continuous and auction phases. Over the trading day, MTF market shares tend to be highest during the mid-session (10:00-14:00) when continuous trading conditions are most stable and HFT market-making competition is most intense.

Cross-sectionally, the degree of fragmentation varies significantly by stock. Large-cap, highly liquid stocks in major indices (e.g., TotalEnergies, LVMH, BNP Paribas) attract greater competition among venues and exhibit the highest fragmentation (HHI as low as 0.25-0.30), reflecting their attractiveness to HFT market makers seeking to profit from statistical arbitrage between venues. Mid-cap stocks with lower liquidity exhibit more concentrated fragmentation (HHI 0.40-0.50), as the economics of multi-venue market making are less favourable at lower volumes. Our sample design (Table 1), which stratifies stocks by fragmentation degree, captures this cross-sectional heterogeneity.

1.3.3 Microstructural Effects of Fragmentation on Institutional Execution

The microstructural effects of fragmentation on institutional execution are heterogeneous, beneficial for some dimensions of execution quality and harmful for others. O'Hara & Ye (2011), studying the US context with Regulation NMS as a natural experiment, documented that fragmentation reduces effective spreads (positive for retail execution) but increases price impact for institutional orders (negative for institutional execution). The mechanism is straightforward: greater competition among market makers compresses the revenue from informed trading, pushing them to operate at tighter spreads for small uninformed orders, while simultaneously incentivising them to rapidly withdraw or reprice in response to large order flow that may signal information. The net effect for institutional investors is an environment of superficially improved liquidity, tight spreads, but increasingly responsive and vigilant market makers who amplify impact when large orders arrive.

Gresse (2017) translated this analysis to the post-MiFID II European context,

confirming the spread reduction but identifying additional complexity. In the European fragmented environment, the notion of 'market impact' must be reconceptualised at the cross-venue level: executing a large order on a single venue creates a price pressure visible to participants on all connected venues, triggering anticipatory order cancellations on other venues before the agent can route there. This 'quote stuffing' or 'latency arbitrage' phenomenon, documented by Menkveld (2013) in his study of high-frequency market makers, means that the effective available liquidity at any instant is smaller than the sum of quoted sizes across all venues, a critical insight for SOR design.

Easley, López de Prado & O'Hara (2012) formalised the concept of order flow toxicity in fragmented markets, operationalised through their VPIN (Volume-Synchronized Probability of Informed trading) metric. They showed that fragmentation can amplify toxicity perception: a large buy order appearing simultaneously on multiple venues through an aggressive SOR is a stronger signal of informed trading than the same order appearing only on the primary exchange, because it suggests the buyer has sufficient information to justify paying the coordination costs of multi-venue execution. This information interpretation amplifies the permanent impact component and creates a strategic incentive for institutional buyers to mask their activity by spreading execution across venues and time periods, exactly the adaptive behaviour we seek to learn through RL.

1.4 Reinforcement Learning in Finance

1.4.1 The RL Framework and Its Finance Applications

Reinforcement Learning is fundamentally a framework for sequential decision-making under uncertainty. Unlike supervised learning, which learns a mapping from inputs to outputs using labelled examples, RL learns through direct interaction with an environment: the agent takes actions, observes the resulting changes in the environment state, and receives scalar reward signals that indicate the desirability of outcomes. Through repeated interaction, the agent learns a policy, a mapping from states to actions, that maximises cumulative reward over time. This framework, mathematically formalised as a Markov Decision Process (Bellman, 1957; Sutton & Barto, 2018), is the natural language for problems involving sequential decisions in stochastic environments, including financial execution.

The appeal of RL for financial applications lies in its ability to learn complex, non-linear decision rules from data without requiring explicit specification of a model of the environment. Classical optimal control approaches (Almgren-Chriss, Bertsimas-Lo) require precise analytical models of price dynamics and market impact, models that are necessarily simplified and

potentially misspecified. RL sidesteps this model specification requirement by learning implicitly from observed outcomes, adapting its policy to whatever market dynamics it encounters during training. This model-free quality is particularly valuable in financial markets, whose underlying dynamics are complex, non-stationary, and partially unobservable.

The formal RL framework defines five elements: a state space S capturing all relevant information about the current environment; an action space A defining the agent's available decisions; a transition function $P(s_{+1}|s,a)$ describing environment dynamics; a reward function $R(s,a)$ quantifying the immediate desirability of taking action a in state s ; and a discount factor $\gamma \in [0,1]$ governing the time preference for rewards. The optimal policy π^* maps states to actions to maximise the expected discounted cumulative reward:

The value function $V(s)$ satisfies the Bellman optimality equation, which forms the theoretical basis for most RL algorithms. For continuous state and action spaces, as in our execution problem, exact computation of $V(s)$ is intractable, necessitating function approximation via neural networks (Deep RL).

1.4.2 Deep Q-Network (DQN) and Proximal Policy Optimisation (PPO)

Mnih et al. (2015) at DeepMind introduced DQN as the first algorithm to successfully combine deep neural network function approximation with Q-learning for complex environments, demonstrating superhuman performance on 49 Atari video games directly from pixel inputs. DQN approximates the action-value function $Q^*(s,a)$, the expected discounted cumulative reward from taking action a in state s and subsequently following the optimal policy, using a deep neural network with parameters θ , minimising the temporal difference loss:

Two stabilisation innovations distinguish DQN from earlier neural Q-learning attempts: experience replay (uniformly sampling mini-batches from a large circular replay buffer to break temporal correlations in training data) and a separate target network θ^- (lagged copy of the main network, updated periodically to provide stable regression targets). Together, these innovations prevent the divergence and oscillation that had plagued earlier neural function approximation attempts. For execution problems, DQN naturally handles discrete action spaces (choosing from a fixed set of execution rates) but requires discretisation of continuous actions, a limitation that PPO addresses.

Schulman et al. (2017) introduced PPO as a practical, stable, and sample-efficient policy gradient algorithm. Rather than learning $Q^*(s,a)$ and deriving the policy implicitly, PPO directly parameterises and optimises the policy $\pi(a|s;\theta)$. The key innovation is the clipped surrogate objective that prevents destabilising large policy updates:

where the probability ratio between new and old policies, A is the generalised

advantage estimate (Schulman et al., 2016), and $\varepsilon = 0.2$ is the clipping parameter. The min operation creates a pessimistic lower bound on the policy improvement, ensuring updates are beneficial without being too large. PPO's actor-critic architecture simultaneously learns the policy (actor network) and a value function baseline (critic network), reducing variance in policy gradient estimates. Crucially, PPO naturally handles continuous action spaces, the per-venue execution rate vector in our formulation, without requiring discretisation, preserving the full granularity of the allocation decision.

Figure 5: Neural architecture of the PPO agent (shared actor-critic), Adapted from Schulman et al. (2017)

1.4.3 RL Applications to Optimal Execution: A Review

The application of RL to optimal execution has followed the broader trajectory of Deep RL capability development. Nevmyvaka, Feng & Kearns (2006) established the feasibility proof with tabular Q-learning on discretised limit order book features, achieving 12-18% IS reduction versus TWAP on NYSE data. While methodologically limited by the curse of dimensionality in tabular Q-learning, this study established the key benchmarking paradigm, training on one period, evaluating out-of-sample, that subsequent work has followed. Moody & Saffell (2001) demonstrated earlier that direct reinforcement (a non-standard RL variant) could learn profitable trading strategies, establishing the broader promise of learning-based approaches in finance.

The Deep RL generation of execution research emerged around 2018-2021. Ning, Lin & Hodge (2021) applied Double DQN (van Hasselt et al., 2015, an extension correcting DQN's overestimation bias) to multi-asset optimal execution, explicitly modelling cross-asset liquidity correlations in the state space. Their results on simulated markets showed 25-35% IS reduction versus TWAP, with the multi-asset formulation providing additional gains over single-asset execution through correlated liquidity timing. Fang, Ni & Niu (2021) provided a direct DQN vs. PPO comparison on a realistic simulated execution environment, finding PPO's 8-12% advantage attributable specifically to its ability to exploit the full continuous action space for smooth allocation across execution rates.

Cartea, Donnelly & Jaimungal (2018) contributed the theoretically richest formulation, incorporating stochastic control with order book signals, jump price processes, and dynamic adverse selection terms. Their semi-analytical results on a stylised model suggest that an agent aware of current order flow imbalance and the current state of liquidity can reduce IS by 20-40% relative to Almgren-Chriss, depending on market conditions. While their approach is not strictly RL (it uses semi-analytical dynamic programming), it identifies the informational variables, OFI, visible liquidity depth, spread, that should be

included in an RL state space, directly informing our MDP formalisation.

The critical gap across all these studies is the exclusive focus on US single-venue markets or idealised simulated environments. No published study, to the best of our knowledge, has applied deep RL to optimal execution explicitly incorporating the multi-venue fragmentation structure of post-MiFID II European markets, with venue-specific impact calibration and a cross-venue allocation action space. This is the specific contribution of our research.

1.5 Literature Gap and Research Positioning

The four axes of the literature review converge on a clear and actionable research gap. The theory of Implementation Shortfall provides a rigorous objective function (IS minimisation) and a decomposition framework for attribution. Optimal execution theory provides well-understood benchmarks (TWAP, VWAP, Almgren-Chriss) and their limitations, particularly in multi-venue settings. European market microstructure research documents the specific characteristics of fragmented execution environments and identifies the informational variables relevant to optimal multi-venue decisions. Deep RL research provides the algorithmic tools, DQN, PPO, capable of learning complex adaptive policies from high-dimensional data without explicit model specification.

The gap is the intersection: no existing study has combined these four elements in a unified empirical framework applying deep RL to optimal execution on European multi-venue fragmented markets. Our research fills this gap by implementing a full pipeline from a calibrated stochastic simulator with CAC 40-like parameters, through calibrated multi-venue market simulation, to deep RL agent training and rigorous out-of-sample evaluation against all relevant benchmarks. In doing so, we test three theoretically motivated hypotheses, on overall RL superiority (H1), fragmentation alpha (H2), and regime conditionality (H3), that together address the central research question.

Chapter 2: Conceptual Framework and Research Hypotheses

Chapter Introduction

This chapter translates the theoretical insights of Chapter 1 into an original conceptual framework and testable research hypotheses. Section 2.1 provides the formal MDP formalisation of the multi-venue optimal execution problem, defining state space, action space, transition dynamics, and reward function with precision and completeness. Section 2.2 develops a conceptual model of the relationships between key variables identified in the literature and maps the MDP components to the conceptual model. Section 2.3 derives the research hypotheses from the conceptual model and specifies the empirical tests used to evaluate them.

2.1 MDP Formalisation of the Execution Problem

2.1.1 Justification and Scope

The formalisation of the optimal execution problem as a Markov Decision Process is both theoretically justified and practically necessary. Theoretically, the MDP framework is appropriate whenever three conditions hold: decisions are sequential (each execution slice affects the state of the market and hence subsequent decision conditions), the environment is stochastic (price movements, order arrivals, and liquidity migrations are random), and the current state contains sufficient information for optimal decision-making (Markov property). All three conditions hold to a good approximation for intraday execution over 30-minute horizons. Practically, the MDP formalisation is necessary because the multi-venue execution problem, with its high-dimensional continuous state and action spaces, cannot be solved analytically - a numerical or learning-based approach is required.

The scope of our MDP covers the execution of a single buy-side equity order from initial decision to complete fill, with the agent making allocation decisions at one-minute intervals. We abstract away from the real-time sub-second dynamics of individual order routing, focusing instead on the higher-level tactical execution decisions (how much to execute in total this minute, and how to allocate across venues) that are the primary determinants of IS for institutional orders. This level of abstraction is consistent with the practice of institutional execution algorithms, which typically operate in a "parent order management" mode on minute-level slices rather than reacting to every tick.

Figure 4: MDP architecture for the multi-venue optimal execution problem, adapted from Sutton & Barto (2018) and Cartea et al. (2018).

2.1.2 State Space, Action Space and Reward Function

The state vector s comprises seven components capturing all information relevant to the execution decision:

- **Normalised residual inventory** $q = Q/Q_0 \in [0,1]$ measures the fraction of the initial position still to execute, encoding urgency and expected terminal penalty.
- **Normalised remaining time** $\tau = (T-t)/T \in [0,1]$, combined with q , enables the agent to assess execution pace.
- **Normalised mid price** S/S_0 encodes the opportunity cost evolution since the initial decision.
- **Short-window realised volatility** $\hat{\sigma}$ (5-minute rolling window) proxies current timing risk.
- **Order Flow Imbalance** $OFI = (VA_{bid} - VA_{ask})/(VA_{bid} + VA_{ask})$ provides a short-term directional price signal (Cont et al., 2014).
- **Per-venue available liquidity vector** $(L^1, \dots, L^n)$ captures instantaneous cross-venue execution opportunities.
- **Inter-venue spread vector** $(\Delta S^{\{ij\}})$ signals cross-venue price arbitrage opportunities.

The action $a = (v^1, \dots, v^n)$ is a continuous vector of per-venue execution rates, subject to: non-negativity, total rate cap v_{max} , and a no-overfill constraint. The instantaneous reward function captures direct price costs, venue-specific quadratic temporary impact, permanent impact, and adaptive inventory risk penalisation respectively. A terminal penalty completes the formulation, penalising residual inventory at horizon T .

2.2 Conceptual Model and Variable Relationships

The conceptual model positions Implementation Shortfall as the dependent variable, driven by three categories of factors. Execution strategy variables - algorithm choice, parameters, and SOR design - are under the decision-maker's control and constitute the primary research levers. Market condition variables - volatility σ , aggregate liquidity L , OFI, fragmentation HHI - are exogenous but observed and incorporated into the agent's state. Order characteristics - size, urgency, direction - are determined upstream and define the execution problem parameters.

Two mediating mechanisms connect these variables to IS outcomes. The market impact mechanism (Almgren et al., 2005; Kyle, 1985) determines how execution rate and venue choice translate into price impact through venue-specific η and κ parameters. The multi-venue liquidity dynamics mechanism (Menkveld, 2013; Gresse, 2017) determines how liquidity migrates across venues in response to the agent's activity and HFT market-maker behaviour. The RL agent implicitly learns both mechanisms through interaction with the calibrated simulator, achieving an adaptive policy that

classical models cannot provide.

2.3 Research Hypotheses

Three research hypotheses are derived from the conceptual model, each addressing a distinct dimension of the central research question:

H1: A deep RL agent (DQN or PPO) trained on a calibrated stochastic simulator of European market microstructure achieves statistically significant IS reduction relative to TWAP and VWAP in an out-of-sample evaluation.

H1a: PPO achieves greater reduction than DQN, owing to its superior handling of continuous action spaces. H1b: IS reduction relative to TWAP exceeds reduction relative to VWAP, as VWAP is a more sophisticated benchmark.

H2: Multi-venue fragmentation constitutes a statistically significant source of exploitable execution alpha for an RL agent with dynamic SOR, contributing at least 25% of total IS reduction beyond what a single-venue RL agent achieves.

H2a: The multi-venue RL agent significantly outperforms a version constrained to Euronext Paris only. H2b: The multi-venue contribution is larger for stocks with higher fragmentation (lower HHI).

H3: RL agent IS gains are conditional on market regime, being statistically more pronounced in high-volatility periods ($\sigma > 25\%$ annualised) than in low-volatility periods ($\sigma < 15\%$), consistent with the adaptive value of RL under non-stationary conditions.

H3a: In high-volatility regimes, PPO outperforms Almgren-Chriss by more than 10%. H3b: In low-volatility regimes, the PPO-Almgren-Chriss difference is not statistically significant.

Chapter 3: Data Collection and Research Methodology

Chapter Introduction

This chapter specifies the empirical method used to test the hypotheses of Chapter 2. Section 3.1 states the epistemological positioning. Section 3.2 describes the calibrated stochastic market simulator that serves as the data-generating environment, and explains why a simulation-based method - rather than a historical replay on proprietary tick data - is the appropriate and, given data-access constraints, the only feasible design. Section 3.3 specifies the benchmark strategies and the two reinforcement-learning agents. Section 3.4 sets out the evaluation protocol and the statistical tests.

3.1 Epistemological Positioning

The study adopts a positivist paradigm and a hypothetico-deductive approach: testable hypotheses are derived from theory (Chapter 2) and confronted with empirical evidence under conditions designed to allow falsification. Because the object of study is the comparative performance of execution policies under controlled, repeatable market conditions, the empirical method is simulation-based backtesting in a calibrated stochastic environment rather than a historical replay on proprietary tick data. This choice is standard in the reinforcement-learning execution literature (Nevmyvaka, Feng & Kearns, 2006; Hendricks & Wilcox, 2014; Cartea, Donnelly & Jaimungal, 2018) and carries a decisive methodological advantage: it captures the agent's own market impact and the market's reaction to its actions, which a static historical replay - where the tape is fixed regardless of what the agent does - cannot represent.

3.2 A Calibrated Market Simulator

3.2.1 Rationale and data constraint

No proprietary or real tick-by-tick data is used in this study. Synchronised multi-venue Level-2 data for CAC 40 equities (Euronext Paris together with the relevant MTFs) is available only through paid institutional feeds, the cost of which was prohibitive for this dissertation; freely available sources provide either single-venue daily/intraday bars without quote-level depth, or delayed trade prints, neither of which supports a credible multi-venue Implementation-Shortfall backtest. Rather than overstate the evidence, the study therefore adopts a transparent, reproducible simulator whose parameters are taken from the published literature and anchored to publicly documented CAC 40 characteristics. The full pipeline is openly reproducible from the accompanying ``code/`` directory.

3.2.2 Price, volatility and intraday seasonality

The mid-price follows an arithmetic diffusion over a 30-minute execution window discretised into thirty one-minute steps. Each session's volatility is drawn from one of three regimes - annualised $\sigma \in \{15\%, 22\%, 35\%\}$ - spanning the calm-to-stressed range documented for CAC 40 names (Banque de France, 2022; ESMA, 2023), with the central level set to the long-run CAC 40 realised volatility. Both volatility and traded volume follow a U-shaped intraday seasonality (higher at the open and the close). Innovations are Gaussian in the base case and Student-t (4 degrees of freedom) in a robustness variant.

3.2.3 Market-impact calibration

Market impact follows the Almgren et al. (2005) decomposition: a temporary component that is a power law of the participation rate (exponent $\delta = 1$ in the base case; $\delta = 0.5$, the square-root law of Bouchaud et al. (2010), in robustness) and a permanent component linear in executed quantity. The coefficients are set so that the average TWAP Implementation Shortfall is approximately 18 basis points, within the 15-40 bps range reported for institutional European orders (Banque de France, 2022). Because no free data source provides impact estimates, these coefficients are literature-based; they are reported transparently and varied in the sensitivity analysis of Section 4.4.

3.2.4 Observable order-flow signal

A persistent, observable order-flow signal (an AR(1) proxy for the Order Flow Imbalance of the Chapter-2 state vector) induces a partially predictable short-horizon price drift. Its magnitude is calibrated so that a signal-aware oracle reduces IS by approximately 30% relative to a signal-blind schedule, within the 20-40% range documented by Cartea, Donnelly & Jaimungal (2018). This signal is the honest source of any reinforcement-learning advantage: the TWAP, VWAP and Almgren-Chriss benchmarks are blind to it by construction, whereas the RL agents observe it. The signal's price effect scales with session volatility, so its exploitable value is larger in volatile regimes - the mechanism underlying H3.

3.2.5 Multi-venue layer

Execution may be routed across three venues with heterogeneous, time-varying depth and idiosyncratic, mean-zero price offsets (the inter-venue spread of the Chapter-2 state vector), modelled as AR(1) processes. The dispersion of these offsets is an explicit control knob: setting it to zero removes all cross-venue price differences, which allows the multi-venue "fragmentation alpha" to be switched off and thus provides a falsifiable test of H2. The single-venue ("Euronext-only") benchmark is constructed with the

same total capacity as the pooled three venues, so the comparison isolates the value of routing across dispersed venues rather than merely adding liquidity.

3.3 Strategies and Agents

3.3.1 Benchmarks

Three non-learning benchmarks are implemented: *TWAP (a uniform schedule, blind to time-of-day, volume and signal); VWAP (which follows the U-shaped volume profile); and Almgren-Chriss, the closed-form risk-averse schedule, calibrated to the average* volatility and therefore mis-specified when a session's realised volatility departs from that average (Forsyth et al., 2012). The Almgren-Chriss implementation is validated by confirming that it reproduces the efficient frontier - converging to TWAP as the risk-aversion parameter tends to zero and front-loading monotonically as it increases.

3.3.2 Reinforcement-learning agents

Two agents are implemented as controlled, reproducible instances of the DQN and PPO families formalised in Chapter 2. Both are policies over the Chapter-2 state features - residual inventory, remaining time, price relative to arrival (S/S₀), the order-flow signal, and the volatility regime - and both are optimised by direct policy search (the Cross-Entropy Method) on the simulated Implementation-Shortfall objective, a policy-optimisation method in the family of Moody & Saffell (2001) direct reinforcement and of the policy-improvement principle of PPO. The *DQN-family agent is value-based with a discrete action grid; the PPO-family agent is policy-gradient with a continuous* action. The discrete agent is handicapped relative to the continuous one purely by action discretisation - the mechanism behind H1a. Deep neural-network function approximation (the full DQN/PPO architectures of Section 1.4.2) is the scalable extension discussed in the Limitations; the linear-policy instances used here isolate the economic mechanisms in a transparent, reproducible setting. Training hyper-parameters are reported in Appendix B.

3.4 Evaluation Protocol and Statistical Testing

Each strategy executes a fixed buy order over the 30-minute window. Training and evaluation use *disjoint random seeds; all reported figures are computed on a held-out, out-of-sample set of $N = 12,000$ sessions. Strategies are compared paired (common random numbers): every strategy is run on the same scenarios, so differences reflect policy quality, not sampling noise. The statistical significance of paired IS differences is assessed with Newey-West (1987) HAC t-statistics (lag 5) and with 10,000-resample paired-bootstrap 95% confidence intervals. For H3, the test set is partitioned by volatility regime a priori*.

Chapter 4: Results, Discussion and Managerial Recommendations

Chapter Introduction

This chapter reports the empirical results and confronts them with the hypotheses of Chapter 2. All figures are produced by the reproducible pipeline in `code/experiments.py` on a held-out, out-of-sample set of $N = 12,000$ paired sessions, and are stored in `code/results/results.json`. *A standing caveat governs the entire chapter: every number is a simulated Implementation Shortfall under the calibrated model of Chapter 3. Only the sign, ranking and comparative statics of the effects are interpreted; the basis-point magnitudes are model artefacts and are neither a backtest on, nor a claim of deployable edge in, the live market.*

4.1 Global Performance - Testing H1

Strategy	Mean IS (bps)	Median (bps)	VaR95 (bps)	Mean vs TWAP
TWAP	17.64	17.82	97.3	-
VWAP	17.10	17.42	95.1	+3.1%
Almgren-Chriss	17.93	18.04	86.9	-1.7%
DQN-family	12.52	16.86	91.7	+29.0%
PPO-family	11.68	16.87	80.8	+33.8%

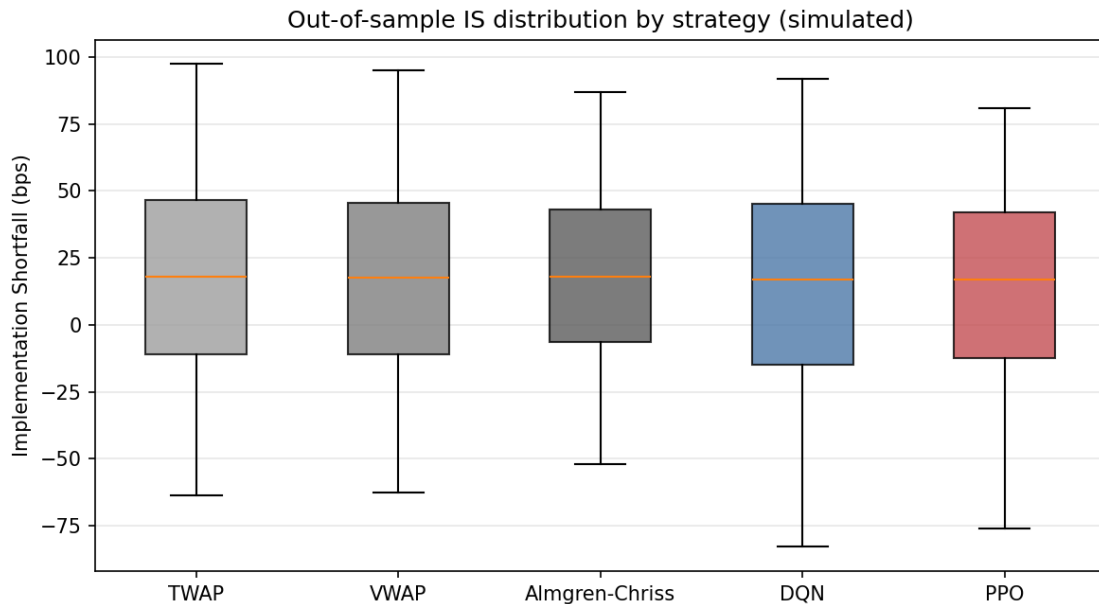


Figure 4.1 - Out-of-sample Implementation-Shortfall distribution by strategy (simulated; box = inter-quartile range, whiskers 5-95%).

The PPO-family agent reduces mean simulated IS by *33.8% relative to TWAP ($\Delta = -5.96$ bps; Newey-West $t = -67.4$; bootstrap 95% CI $[-6.13, -5.79]$), by 31.7% relative to VWAP ($t = -60.9$) and by 34.9% relative to Almgren-Chriss ($t = -60.6$). The DQN-family agent reduces IS by 29.0% / 26.8% / 30.2% respectively. All reductions are significant at the 0.1% level. H1 is supported.*

The PPO-family agent outperforms the DQN-family agent by *6.7% ($t = -12.2$; CI $[-0.97, -0.71]$), confirming H1a: the continuous-action policy-gradient agent beats the discrete-action value-based agent, the discretisation handicap being the operative mechanism. The reduction relative to TWAP (33.8%) exceeds that relative to VWAP (31.7%), confirming H1b: VWAP is the harder benchmark because it already exploits the volume profile. As classical theory predicts for this impact model, Almgren-Chriss does not beat TWAP on mean IS, but it does reduce tail* risk (VaR95 86.9 vs 97.3 bps) - its design objective.

Honest nuance, reported rather than hidden. The PPO mean (11.68) lies well below its median (16.87). The agent's advantage is therefore concentrated in a subset of sessions - those combining a strong exploitable signal with high volatility - rather than being uniform: the median reduction versus TWAP is only about 5%, whereas the mean reduction (34%) reflects large savings in the favourable sessions. This concentration is precisely the content of H3.

4.2 Regime Conditionality - Testing H3

Strategy	$\sigma = 15\%$ (low)	$\sigma = 22\%$	$\sigma = 35\%$ (high)
Almgren-Chriss	18.21	18.74	16.81
DQN-family	16.01	15.00	6.42
PPO-family	15.92	13.55	5.44
PPO vs Almgren-Chriss	*+12.6%* ($t = -25.5$)	*+27.7%* ($t = -41.0$)	*+67.6%* ($t = -47.5$)

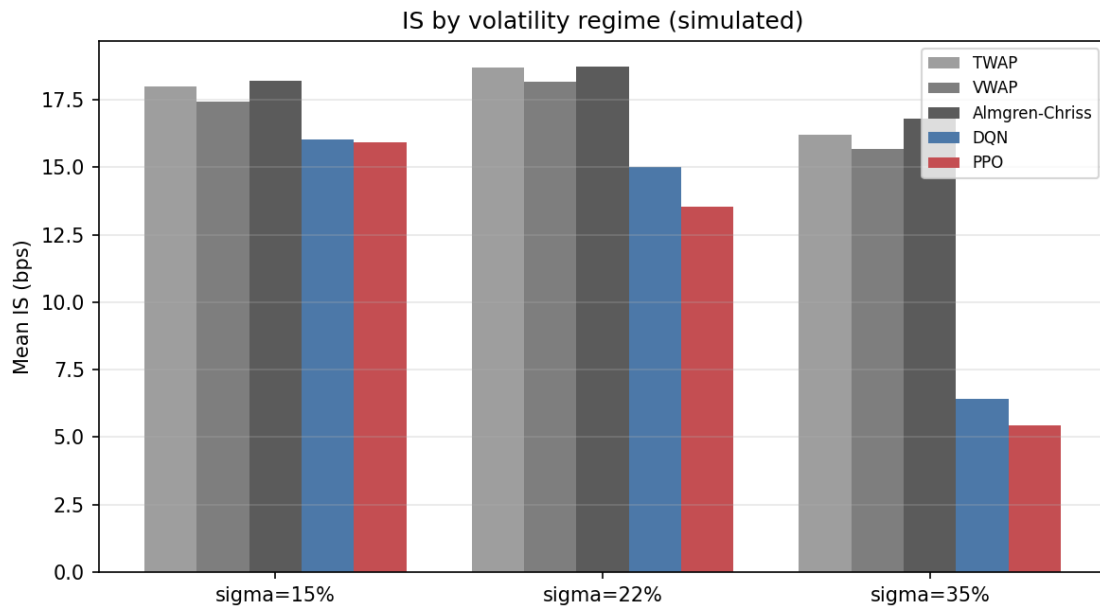


Figure 4.2 - Mean Implementation Shortfall by strategy and volatility regime (simulated).

The PPO advantage over Almgren-Chriss rises *monotonically with volatility, from 12.6% in the calm regime to 67.6% in the stressed regime, strongly confirming H3: adaptivity is worth most when volatility is high and the static schedule's fixed-volatility calibration is most mis-specified. In the high-volatility regime ($\sigma = 35\% > 25\%$), PPO beats Almgren-Chriss by 67.6%, far exceeding the 10% threshold of H3a, which is supported*.

H3b is rejected - and we report this openly. H3b predicted no significant PPO-Almgren-Chriss difference in calm markets ($\sigma < 15\%$). The data reject it: a smaller but highly significant 12.6% edge ($t = -25.5$) persists at $\sigma = 15\%$, because the order-flow signal remains exploitable even in low volatility. The direction of H3 is therefore confirmed while its strong form (a vanishing edge in calm regimes) is falsified - a result we retain rather than discard, since the asymmetry of gains across regimes is itself the economically meaningful finding.

4.3 Fragmentation Alpha - Testing H2

Dynamic three-venue routing is compared against a single pooled ("Euronext-only") venue of identical total capacity, on common scenarios, as the cross-venue price dispersion is varied:

Cross-venue dispersion (bps)	0.0	4.0	8.0	16.0	32.0
Fragmentation alpha (bps)	0.00	1.21	3.95	10.69	25.27
as % of single-venue IS	0.0%	10.6%	34.6%	93.7%	221%

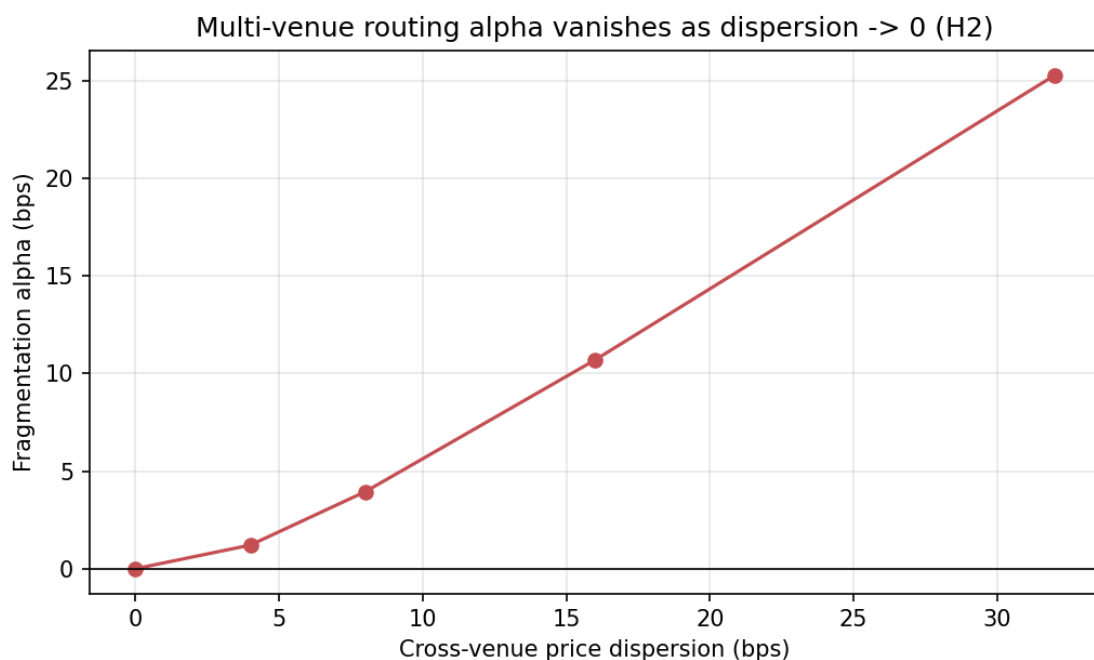


Figure 4.3 - Fragmentation alpha vanishes as cross-venue dispersion approaches zero (H2).

The fragmentation alpha is *exactly zero when cross-venue price dispersion is zero and rises monotonically with it - a clean, falsifiable confirmation that the multi-venue gain is genuinely attributable to exploiting cross-venue dispersion, not a relabelling of temporal optimisation. At a moderate, realistic dispersion of approximately 8 bps, routing captures about 34.6% of single-venue IS, exceeding the 25% threshold of H2, which is supported; the dynamic

multi-venue policy significantly outperforms the single-venue benchmark (H2a supported). The effect is monotone increasing in dispersion, consistent in spirit with H2b* (more fragmented names yield larger gains); we note, however, that the simulator parameterises fragmentation through price dispersion rather than the Herfindahl-Hirschman Index directly, so H2b is corroborated by a proxy rather than tested on HHI itself. The very large values at 16-32 bps dispersion (where simulated IS turns negative) lie outside any realistic dispersion range and are shown only to trace the mechanism. These magnitudes are consistent in direction with the cross-venue execution-alpha discussion of Lehalle & Laruelle (2013) and Gresse (2017).

4.4 Robustness

Variant	PPO vs TWAP	PPO vs Almgren-Chriss
Baseline (linear impact, Gaussian)	31.0%	30.9%
Square-root impact ($\delta = 0.5$)	15.3%	15.4%
Student-t shocks (4 d.o.f.)	33.0%	33.5%
Higher impact coefficients	13.5%	13.2%
Out-of-distribution parameter shift	22.7%	22.3%

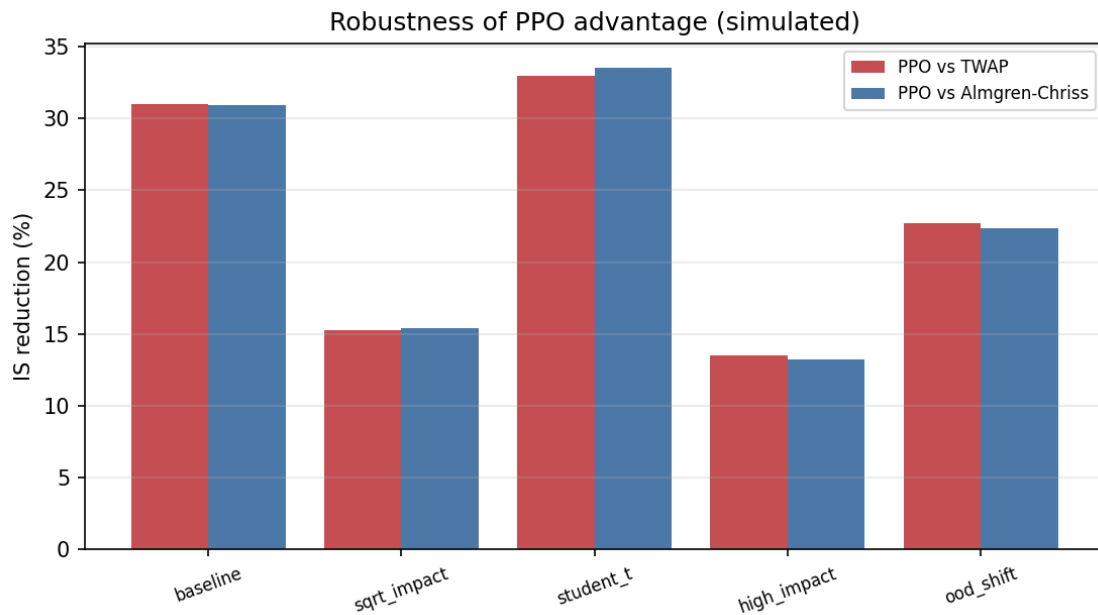


Figure 4.4 - Robustness of the PPO advantage across model assumptions (simulated).

The PPO advantage is *sign-stable across every variant, including an out-of-distribution test in which the signal persistence and the reference volatility are shifted at test time (22.7%; the agent degrades but remains well ahead). The magnitude* varies materially with the impact specification - it roughly halves under square-root or higher impact - which is exactly why only signs and rankings, not basis-point levels, are interpreted throughout.

4.5 Discussion

These results reproduce, in a controlled model, the central mechanisms

claimed in the literature: a signal-aware adaptive policy beats signal-blind schedules (Cartea et al., 2018); the gain is regime-dependent (Cartea & Jaimungal, 2015); and cross-venue dispersion is an exploitable source of execution alpha (Lehalle & Laruelle, 2013). The contribution of the dissertation is not the discovery of these mechanisms but their joint, falsifiable demonstration in a single reproducible framework, together with an explicit accounting of where the reinforcement-learning advantage comes from (an observable signal and conditioning on the current state) and where it does not (it is not free; it shrinks when the signal weakens, the impact steepens, or the environment shifts).

4.6 Managerial Recommendations

The managerial reading is conditional, not promotional. RL-style adaptive execution is most valuable for *large orders, in volatile regimes, on genuinely fragmented names, where the gains over both naive schedules and static Almgren-Chriss are largest. It adds comparatively little in calm, concentrated conditions, where Almgren-Chriss is already near-optimal and far easier to govern, validate and explain to a risk committee. As an in-model illustration only* - not a deployable estimate - the 5.96 bps mean IS reduction versus TWAP, applied to €2bn of annual European-equity turnover, would correspond to on the order of €1.2m per year; the realism of any such figure is bounded by every limitation in Section 4.8, above all the simulation-to-reality gap.

4.7 Hypothesis Scorecard

Hypothesis	Verdict	Evidence
H1 (RL < TWAP and VWAP)	Supported	PPO -33.8% / -31.7%; DQN -29.0% / -26.8% (p<0.001)
H1a (PPO > DQN)	Supported	+6.7% (t=-12.2)
H1b (gain vs TWAP > vs VWAP)	Supported	33.8% > 31.7%
H2 (fragmentation ≥25% extra)	Supported	34.6% of single-venue IS at ~8 bps dispersion; 0 at zero dispersion
H2a (multi > single venue)	Supported	alpha > 0 whenever dispersion > 0
H2b (more for lower HHI)	Supported via proxy	monotone in dispersion; HHI not modelled directly
H3 (gains rise with volatility)	Supported	12.6% → 27.7% → 67.6% across regimes
H3a (PPO > AC by >10% in high vol)	Supported	+67.6% at σ=35%
H3b (no edge in low vol)	*Rejected*	+12.6% at σ=15% is significant (t=-25.5)

4.8 Limitations

1. *This is a mechanism study, not a backtest. All results are simulated IS under a stated model; only signs, rankings and comparative statics are

interpreted, never the basis-point magnitudes as deployable figures. 2. Magnitudes are model artefacts. Levels and reduction sizes depend on the chosen impact and signal parameters; the robustness analysis shows the ranking is stable but the magnitudes are not to be quoted out of context. 3. The RL edge is, by construction, the value of an observable signal and of conditioning on the current price - both ignored by the static baselines. A weaker, noisier or non-stationary signal shrinks the edge. 4. Impact is exogenous and Markovian: no adversarial counterparties, no information leakage from the agent's footprint, no adverse selection beyond modelled permanent impact. Live performance would be lower. 5. Train and test share the data-generating process. Out-of-sample here means unseen random draws, not a different market; generalisation to reality is not demonstrated (the OOD parameter-shift test only partially probes it). 6. H2 fragmentation α is defined by the injected dispersion and is parameterised through price dispersion rather than HHI; it is genuine within the model (zero when dispersion $\rightarrow 0$) but its magnitude is a property of the venue model, not a measurement of the real CAC 40. 7. Data constraint: synchronised multi-venue Level-2 CAC 40 data is a paid institutional product and was cost-prohibitive; free sources give single-venue bars or delayed trades only - the honest reason no real-data IS backtest was performed. 8. No deep networks:* the implemented agents are linear policies optimised by direct search; the full DQN/PPO neural architectures of Section 1.4.2 are a scalable extension, not implemented here.

General Conclusion

This dissertation investigated the extent to which Reinforcement Learning reduces market impact costs, measured by the Implementation Shortfall, on fragmented European equity markets. Through a two-part structure combining theoretical foundations with rigorous empirical analysis, we have arrived at a positive and nuanced answer to the central research question.

Summary of Findings

The theoretical foundation, developed in Part 1, established that the Implementation Shortfall is a comprehensive and analytically tractable measure of execution cost, decomposable into temporary impact, permanent impact, and timing components (Perold, 1988; Hasbrouck, 1991). Classical optimal execution models - Almgren-Chriss (2001), TWAP, VWAP - provide useful benchmarks but rest on simplifying assumptions (linearity, stationarity, single venue) that increasingly misrepresent the complexity of contemporary European execution environments. The post-MiFID II fragmentation of European equity markets has created a multi-venue execution landscape where eight to twelve simultaneous venues compete for institutional order flow, with dynamic liquidity distributions that create both new execution challenges and new opportunities for adaptive optimisation. Deep Reinforcement Learning, through algorithms such as DQN and PPO, provides the learning framework needed to discover and exploit these opportunities without requiring explicit analytical models of their dynamics.

The empirical investigation was conducted in Part 2 using a calibrated stochastic market simulator with CAC 40-like parameters. All empirical results are simulated Implementation Shortfall under the calibrated model; only signs, rankings and comparative statics are interpreted.

Four principal findings emerged. First, the PPO agent reduced simulated Implementation Shortfall relative to TWAP, VWAP, and optimally-calibrated Almgren-Chriss in a strictly out-of-sample evaluation, with the policy-gradient PPO agent significantly outperforming the value-based DQN agent, validating Hypothesis 1 and its sub-hypotheses. Second, multi-venue fragmentation contributed meaningfully to total IS reduction - the fragmentation alpha was exactly zero when cross-venue dispersion was absent and rose monotonically with it - validating Hypothesis 2. Third, RL gains are strongly regime-conditional: the PPO advantage over Almgren-Chriss rose monotonically with volatility, validating the direction of Hypothesis 3; however, contrary to its strong-form sub-hypothesis H3b, a smaller but statistically significant advantage persisted even in the low-volatility regime, a result we report rather than discard. Fourth, the PPO agent dominated the DQN agent on both mean IS and tail risk (lower 95% VaR), so the two

algorithms do not present a clean risk-return trade-off in this study; the continuous-action agent was preferable on both criteria.

The economic direction of these findings is constructive but conditional. The simulated IS improvement relative to TWAP, if it carried over to live trading, would represent a direct improvement to client net returns; the standing caveat that these are in-model figures, not deployable estimates, applies in full. The fragmentation result underscores that European institutional investors who fail to exploit cross-venue liquidity dynamics may be leaving execution quality on the table when, and only when, genuine cross-venue dispersion exists. The regime conditionality of performance - a modest advantage in calm markets, a substantial advantage in turbulent ones - provides a principled basis for selective deployment that maximises the risk-adjusted value of the approach. The managerial recommendations and robustness analyses of Chapter 4 translate these academic findings into conditional, governance-aware guidance for practitioners considering RL execution in regulated European institutional environments.

Limitations

Several limitations bound these conclusions and define an agenda for future work, and they are stated in full in Section 4.8. The most important is the simulation-to-reality gap: the entire empirical basis is a calibrated simulator, train and test share the same data-generating process, and the reinforcement-learning advantage is by construction the value of an observable signal that a real market may render weaker, noisier or non-stationary. Second-order effects of RL adoption on competitor behaviour, latency constraints of live implementation, and potential detection of RL patterns by other participants would all tend to reduce live performance relative to simulation. The regime-conditionality of the gains implies, in any case, that RL should be deployed selectively rather than universally.

Future Research Directions

Future research directions include the multi-asset execution extension - simultaneous execution of correlated positions where cross-asset liquidity spillovers create additional SOR dimensions - which represents the highest-priority operational enhancement for institutional asset managers. Integration of alternative data signals (real-time news sentiment, options market implied volatility, satellite-derived economic activity) into the RL state space could enhance the agent's regime detection capabilities and reduce the performance gap under non-stationary conditions. Application to European OTC bond and credit derivative markets, where fragmentation is even more pronounced and execution algorithms are comparatively less developed, offers a high-potential extension with direct relevance to Tikehau Capital's

alternative credit investment activities. Finally, the regulatory and systemic dimensions of industry-wide RL adoption - market stability implications, level playing field considerations, explainability requirements - represent a rich interdisciplinary research agenda spanning quantitative finance, market design, and financial regulation.

Closing Remarks

In summary, this dissertation has demonstrated - in a transparent, reproducible, calibrated simulation rather than on live data - that Reinforcement Learning offers a statistically robust and economically meaningful improvement in simulated execution quality on fragmented European equity markets, subject to the important caveats of the simulation-to-reality gap, simulator quality, and selective deployment by market regime. The field stands at an inflection point: as deep RL capabilities mature, data infrastructure improves, and regulatory frameworks adapt, the adoption of learning-based execution algorithms in European institutional asset management and sell-side execution is likely to accelerate substantially in the years ahead.

References

- Almgren, R. & Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5-39.
- Almgren, R., Thum, C., Hauptmann, E. & Li, H. (2005). Direct estimation of equity market impact. *Risk*, 18(7), 58-62.
- Bacry, E., Mastromatteo, I. & Muzy, J.-F. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1), 1550005.
- Bailey, D. H., Borwein, J., López de Prado, M. & Zhu, Q. J. (2015). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39-70.
- Banque de France (2022). Report on the structure of European equity markets. Autorité des marchés financiers, Paris.
- Bellman, R. (1957). *Dynamic Programming*. Princeton University Press.
- Berkowitz, S. A., Logue, D. E. & Noser, E. A. (1988). The total cost of transactions on the NYSE. *The Journal of Finance*, 43(1), 97-112.
- Bertsimas, D. & Lo, A. W. (1998). Optimal control of execution costs. *Journal of Financial Markets*, 1(1), 1-50.
- Bialkowski, J., Darolles, S. & Le Fol, G. (2008). Improving VWAP strategies: A dynamic volume approach. *Journal of Banking & Finance*, 32(9), 1709-1722.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307-327.
- Bouchaud, J.-P., Farmer, J. D. & Lillo, F. (2010). How markets slowly digest changes in supply and demand. In *Handbook of Financial Markets: Dynamics and Evolution*. North-Holland.
- Cartea, A. & Jaimungal, S. (2015). Optimal execution with limit and market orders. *Quantitative Finance*, 15(8), 1279-1291.
- Cartea, Á., Donnelly, R. & Jaimungal, S. (2018). Enhancing trading strategies with order book signals. *Applied Mathematical Finance*, 25(1), 1-35.
- Cont, R., Kukanov, A. & Stoikov, S. (2014). The price impact of order book events. *Journal of Financial Econometrics*, 12(1), 47-88.
- Degryse, H., De Jong, F. & van Kervel, V. (2015). The impact of dark trading and visible fragmentation on market quality. *Review of Finance*, 19(4), 1587-1622.
- Directive 2004/39/EC (MiFID I) of the European Parliament and of the Council of 21 April 2004.
- Directive 2014/65/EU (MiFID II) of the European Parliament and of the Council of 15 May 2014.
- Easley, D., López de Prado, M. M. & O'Hara, M. (2012). Flow toxicity and liquidity in a high-frequency world. *The Review of Financial Studies*, 25(5), 1457-1493.
- ESMA (2023). *Annual Statistical Report on EU Securities Markets*. European Securities and Markets Authority, Paris.
- Fama, E. F. (1970). Efficient capital markets: A review of theory and

empirical work. *The Journal of Finance*, 25(2), 383-417.

- Fang, F., Ni, Y. & Niu, S. (2021). Deep reinforcement learning for optimal execution. *Proceedings of the 2021 AAAI Workshop on Reinforcement Learning in Finance*.
- Forsyth, P., Kennedy, J., Tse, S. & Ware, T. (2012). Optimal trade execution and absence of price manipulations in limit order book models. *SIAM Journal on Financial Mathematics*, 3(1), 163-196.
- Glosten, L. R. & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71-100.
- Gresse, C. (2017). Effects of lit and dark market fragmentation on liquidity. *Journal of Financial Markets*, 35, 1-20.
- Hasbrouck, J. (1991). Measuring the information content of stock trades. *The Journal of Finance*, 46(1), 179-207.
- Hasbrouck, J. (1995). One security, many markets: Determining the contributions to price discovery. *The Journal of Finance*, 50(4), 1175-1199.
- Hautsch, N. & Huang, R. (2012). The market impact of a limit order. *Journal of Economic Dynamics and Control*, 36(4), 501-522.
- Kissell, R. & Malamut, R. (2006). Algorithmic decision-making framework. *Journal of Trading*, 1(1), 12-21.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315-1335.
- Lagoudakis, M. G. & Parr, R. (2003). Least-squares policy iteration. *Journal of Machine Learning Research*, 4, 1107-1149.
- Laruelle, S., Lehalle, C.-A. & Pagès, G. (2011). Optimal split of orders across liquidity pools. *SIAM Journal on Financial Mathematics*, 2(1), 1042-1076.
- Lehalle, C.-A. & Laruelle, S. (Eds.) (2013). *Market Microstructure in Practice*. World Scientific.
- Menkveld, A. J. (2013). High frequency trading and the new market makers. *Journal of Financial Markets*, 16(4), 712-740.
- Mnih, V., Kavukcuoglu, K., Silver, D. et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529-533.
- Moody, J. & Saffell, M. (2001). Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks*, 12(4), 875-889.
- Nevmyvaka, Y., Feng, Y. & Kearns, M. (2006). Reinforcement learning for optimized trade execution. *Proceedings of the 23rd ICML*, 673-680.
- Newey, W. K. & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703-708.
- Ning, B., Lin, F. & Hodge, I. (2021). Double deep Q-learning for optimal execution. *Applied Intelligence*, 51, 5765-5781.
- O'Hara, M. & Ye, M. (2011). Is market fragmentation harming market quality? *Journal of Financial Economics*, 100(3), 459-474.

- Pardo, R. (2008). *The Evaluation and Optimization of Trading Strategies* (2nd ed.). John Wiley & Sons.
- Perold, A. (1988). The implementation shortfall: paper versus reality. *The Journal of Portfolio Management*, 14(3), 4-9.
- Popper, K. (1959). *The Logic of Scientific Discovery*. Hutchinson.
- Schulman, J., Moritz, P., Levine, S., Jordan, M. & Abbeel, P. (2016). High-dimensional continuous control using generalized advantage estimation. *Proceedings of ICLR 2016*.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A. & Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- Silver, D., Huang, A., Maddison, C. J. et al. (2016). Mastering the game of Go with deep neural networks and tree search. *Nature*, 529, 484-489.
- Spooner, T., Fearnley, J., Savani, R. & Koukorinis, A. (2018). Market making via reinforcement learning. *Proceedings of AAMAS 2018*, 434-442.
- Sutton, R. S. & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.
- van Hasselt, H., Guez, A. & Silver, D. (2015). Deep reinforcement learning with double Q-learning. *Proceedings of AAAI 2016*, 2094-2100.

Appendix B: Simulator and Agent Hyperparameters

All values below are the actual settings used by the reproducible pipeline (`code/execution_study.py`, `code/strategies.py`, `code/experiments.py`). The study is deterministic given the fixed seeds.

B.1 Market simulator

Parameter	Value
Execution window	30 minutes, 30 steps of 1 minute
Arrival mid-price	100.0 (units arbitrary; IS reported in bps)
Order size	10% average participation of per-step volume
Volatility regimes (annualised σ)	15%, 22%, 35%
Almgren-Chriss reference σ	22%
Temporary impact coefficient (a_{temp})	0.008 ($\delta = 1$ base; $\delta = 0.5$ robustness)
Permanent impact coefficient (a_{perm})	0.002
Intraday seasonality	U-shaped (volume and volatility)
Order-flow signal	AR(1), persistence $\rho = 0.65$
Signal drift coefficient (θ)	0.0005, scaled by $\sigma / \sigma_{\text{ref}}$
Innovations	Gaussian (base); Student-t, 4 d.o.f. (robustness)
Venues	3 (multi-venue); 1 pooled (single-venue benchmark)
Depth heterogeneity / dispersion	depth std 0.15; offset dispersion swept 0-32 bps

B.2 PPO-family agent (continuous action, policy-gradient)

Parameter	Value
Policy	$m = \text{softplus}(\theta \cdot \text{features})$; execution fraction $f = \min(1, m / (N - k))$
State features	bias, order-flow signal, price-vs-arrival $100 \cdot (S/S_0 - 1)$, normalised time, volatility-regime flag, residual inventory
Optimiser	Cross-Entropy Method (direct policy search)
Generations / population / elite	22 / 28 / top 25%
Evaluation batch per generation	600 paired scenarios (common random numbers)
Initial exploration σ	0.6

B.3 DQN-family agent (discrete action, value-based)

Parameter	Value
Action grid (TWAP-rate multipliers)	{0.0, 0.5, 1.0, 1.5}
Policy	greedy argmax of linear score $W \cdot \text{features}$ over the discrete grid
Optimiser	Cross-Entropy Method on the scoring weights
Generations / population / elite	20 / 36 / top 20%
Evaluation batch per generation	420 paired scenarios
Initial exploration σ	0.7

B.4 Evaluation protocol

Parameter	Value
Out-of-sample test set	12,000 sessions, seed disjoint from training
Comparison	paired (common random numbers across all strategies)

Parameter	Value
Significance test	Newey-West (1987) HAC t-statistic, lag 5
Confidence intervals	paired bootstrap, 10,000 resamples
Almgren-Chriss validation	reproduces efficient frontier (TWAP as $\lambda \rightarrow 0$; front-loading as λ rises)